Deep Neural Network for Estimating m-Height

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1 Background

Let k and n be two positive integers, where $k < n$. Let

$$G = \begin{bmatrix} g_{0,0} & g_{0,1} & \cdots & g_{0,n-1} \\ g_{1,0} & g_{1,1} & \cdots & g_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ g_{k-1,0} & g_{k-1,1} & \cdots & g_{k-1,n-1} \end{bmatrix}$$

be a full-rank matrix of k rows and n columns, where each element $g_{i,j}$ is a real number. (Since here $k < n$, the rank of the matrix G is k. And we assume that no column of G is the all-zero vector, namely, for $j = 0, 1, \cdots, n-1$, $(g_{0,j}, g_{1,j}, \cdots, g_{k-1,j}) \neq (0, 0, \cdots, 0)$.) The matrix G is called *systematic* if its first k columns form an identity matrix. That is,

$$G = \begin{bmatrix} 1 & 0 & \cdots & 0 & g_{0,k} & g_{0,k+1} & \cdots & g_{0,n-1} \\ 0 & 1 & \cdots & 0 & g_{1,k} & g_{1,k+1} & \cdots & g_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & g_{k-1,k} & g_{k-1,k+1} & \cdots & g_{k-1,n-1} \end{bmatrix}$$

In this case, we can use $I_{k\times k}$ to denote the $k \times k$ identity matrix, and use the matrix $P_{k,n-k}$ to denote the last $n - k$ columns of G, then we have

$$G = \left[I_{k\times k} \mid P_{k\times(n-k)} \right]$$

Example 1. *For example, when $k = 2$ and $n = 5$, the matrix G might be*

$$G = \begin{bmatrix} 1 & 0 & 0.4759809 & 0.9938236 & 0.819425 \\ 0 & 1 & -0.8960798 & -0.7442706 & 0.3345122 \end{bmatrix}$$

□

Let

$$\mathbf{x} = (x_0, x_1, \cdots, x_{k-1}) \in \mathbb{R}^k$$

be any vector of length k, where each of the k elements is a real number. (The symbol $\mathbb{R}$ denotes the set of all real numbers. And $\mathbb{R}^k$ denotes the k-dimensional real vector space, namely, the set of all real-valued vectors of length k.) When we multiply the vector $\mathbf{x}$ with the matrix G, we get a real-valued vector

$$\mathbf{c} = (c_0, c_1, \cdots, c_{n-1})$$

of length n:

$$\mathbf{c} = \mathbf{x}G = (x_0, x_1, \cdots, x_{k-1}) \begin{bmatrix} g_{0,0} & g_{0,1} & \cdots & g_{0,n-1} \\ g_{1,0} & g_{1,1} & \cdots & g_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ g_{k-1,0} & g_{k-1,1} & \cdots & g_{k-1,n-1} \end{bmatrix} \in \mathbb{R}^n$$

For $i = 0, 1, \cdots, n-1$, we have

$$c_i = (x_0, x_1, \cdots, x_{k-1}) \cdot \begin{pmatrix} g_{0,i} \\ g_{1,i} \\ \vdots \\ g_{k-1,i} \end{pmatrix} = x_0 g_{0,i} + x_1 g_{1,i} + \cdots + x_{k-1} g_{k-1,i} = \sum_{j=0}^{k-1} x_j g_{j,i}$$

Example 2. *Let G be the matrix in the previous example, where $k = 2$ and $n = 5$. Let $\mathbf{x} = (-1.2, 3.8)$. Then the vector $\mathbf{c} = \mathbf{x}G$ is*

$$\mathbf{c} = (c_0, c_1, c_2, c_3, c_4) = (-1.2, 3.8, -3.97628032, -4.0208166, 0.28783636)$$

□

The vector $\mathbf{c} = \mathbf{x}G \in \mathbb{R}^n$ is called an *analog codeword*, or simply *codeword*. (The word *analog* means "real value" or "continuous value", instead of "discrete value".) The set of all such codewords

$$\mathcal{C} = \{\mathbf{x}G \mid \mathbf{x} \in \mathbb{R}^k\}$$

is called an *Analog Code*, or simply *code*. Note that the code $\mathcal{C}$ is a k-dimensional subspace inside an n-dimensional space. Since this code $\mathcal{C}$ is generated using the matrix G, G is called the *Generator Matrix* of the code $\mathcal{C}$.

Let the function

$$\pi \ : \ \{0, 1, \cdots, n-1\} \to \{0, 1, \cdots, n-1\}$$

be a permutation that maps every integer $i \in \{0, 1, \cdots, n-1\}$ to another integer $\pi(i) \in \{0, 1, \cdots, n-1\}$. A permutation is a bijection (a one-to-one mapping), which implies that no two integers are mapped to the same integer; in other words, for any $i \neq j$, we have $\pi(i) \neq \pi(j)$. Given a codeword $\mathbf{c} = (c_0, c_1, \cdots, c_{n-1}) \in \mathcal{C}$ and a permutation π, we get a permutated vector

$$\mathbf{c}_\pi = (c_{\pi(0)}, c_{\pi(1)}, \cdots, c_{\pi(n-1)})$$

Note that $\mathbf{c}_\pi$ is not necessarily a codeword in $\mathcal{C}$.

Example 3. *Let the codeword be*

$$\mathbf{c} = (c_0, c_1, c_2, c_3, c_4) = (-1.2, 3.8, -3.97628032, -4.0208166, 0.28783636),$$

and let the permutation π be

$$\pi(0) = 3, \ \pi(1) = 2, \ \pi(2) = 1, \ \pi(3) = 0, \ \pi(4) = 4$$

Then

$$\mathbf{c}_\pi = (-4.0208166, -3.97628032, 3.8, -1.2, 0.28783636)$$

□

Given a codeword $\mathbf{c} = (c_0, c_1, \cdots, c_{n-1}) \in \mathcal{C}$ and a permutation π, if

$$|c_{\pi(0)}| \geq |c_{\pi(1)}| \geq \cdots \geq |c_{\pi(n-1)}|$$

then π is called a *Ranking Permutation* of $\mathbf{c}$.

Example 4. *The permutation π in Example 3 is actually a* Ranking Permutation *of $\mathbf{c}$, because*

$$|-4.0208166| \geq |-3.97628032| \geq |3.8| \geq |-1.2| \geq |0.28783636|$$

□

Consider any integer

$$m \in \{0, 1, \cdots, n-1\}.$$

Given a codeword $\mathbf{c} = (c_0, c_1, \cdots, c_{n-1}) \in \mathcal{C}$ and its *Ranking Permutation* π, the "m-height of $\mathbf{c}$" is defined as follows:

- If $\mathbf{c}$ is not the all-zero vector $(0, 0, \cdots, 0)$ and $c_{\pi(m)} \neq 0$, the "m-height of $\mathbf{c}$" is defined as

$$h_m(\mathbf{c}) = \left| \frac{c_{\pi(0)}}{c_{\pi(m)}} \right|$$

- If $\mathbf{c}$ is not the all-zero vector $(0, 0, \cdots, 0)$ and $c_{\pi(m)} = 0$, the "m-height of $\mathbf{c}$" is defined as $h_m(\mathbf{c}) = \infty$.
- If $\mathbf{c}$ is the all-zero vector $(0, 0, \cdots, 0)$, the "m-height of $\mathbf{c}$" is defined as $h_m(\mathbf{c}) = 0$.

We can see that given a codeword $\mathbf{c} = (c_0, c_1, \cdots, c_{n-1}) \in \mathcal{C}$, its m-height sequence

$$h_0(\mathbf{c}),\ h_1(\mathbf{c}),\ \cdots,\ h_{n-1}(\mathbf{c})$$

is determined, and we have

$$h_0(\mathbf{c}) \ \leq \ h_1(\mathbf{c}) \ \leq \ \cdots \ \leq \ h_{n-1}(\mathbf{c}).$$

Example 5. *For the codeword* $\mathbf{c}$ *in Example 3, since*

$$|-4.0208166| \geq |-3.97628032| \geq |3.8| \geq |-1.2| \geq |0.28783636|$$

we have

$$\begin{array}{rcl} h_0(\mathbf{c}) & = & \left| \frac{-4.0208166}{-4.0208166} \right| = 1 \\ h_1(\mathbf{c}) & = & \left| \frac{-4.0208166}{-3.97628032} \right| = 1.0112004879977878 \\ h_2(\mathbf{c}) & = & \left| \frac{-4.0208166}{3.8} \right| = 1.0581096315789473 \\ h_3(\mathbf{c}) & = & \left| \frac{-4.0208166}{-1.2} \right| = 3.3506805 \\ h_4(\mathbf{c}) & = & \left| \frac{-4.0208166}{0.28783636} \right| = 13.969105918376677 \end{array}$$

□

For the analog code $\mathcal{C}$ (which consists of infinitely many codewords), its m-height is defined as

$$h_m(\mathcal{C}) = \max_{\mathbf{c} \in \mathcal{C}} \ h_m(\mathbf{c})$$

that is, the m-height of the code $\mathcal{C}$ is the maximum m-height of all its codewords.

Example 6. *Let* $\mathcal{C}$ *be the code in the previous examples, and let* $m = 2$*. In Example 5, it has been shown that for the codeword* $\mathbf{c} = (c_0, c_1, c_2, c_3, c_4) = (-1.2, 3.8, -3.97628032, -4.0208166, 0.28783636)$*, its 2-height is*

$$h_2(\mathbf{c}) = 1.0581096315789473.$$

So we know that

$$h_2(\mathcal{C}) \geq 1.0581096315789473$$

namely, $h_2(\mathbf{c})$ *(the 2-height of the codeword* $\mathbf{c}$*) gives us a lower bound to* $h_2(\mathcal{C})$ *(the 2-height of the code* $\mathcal{C}$*).*

There exist other codewords in $\mathcal{C}$ *that have greater 2-heights than* $\mathbf{c}$ *does. For example, for the codeword*

$$\begin{array}{rcl} \mathbf{c}^* & = & (-0.435, -1.924) \begin{bmatrix} 1 & 0 & 0.4759809 & 0.9938236 & 0.819425 \\ 0 & 1 & -0.8960798 & -0.7442706 & 0.3345122 \end{bmatrix} \\ & = & (-0.435, -1.924, 1.5170058436999998, 0.9996633684, -1.0000513478) \end{array}$$

When we order the 5 elements of $\mathbf{c}^*$ *by their absolute values (from high to low), they become*

$$|-1.924| \ \geq \ |1.5170058436999998| \ \geq \ |-1.0000513478| \ \geq \ |0.9996633684| \ \geq \ |-0.435|$$

So the 2-height of the codeword $\mathbf{c}^*$ *is*

$$h_2(\mathbf{c}^*) = \left|\frac{-1.924}{-1.0000513478}\right| = 1.9239012119053513$$

It can be shown that the actual 2-height of the code is

$$h_2(\mathcal{C}) = 1.9242387$$

So the 2-height of $\mathbf{c}^*$ *is quite close to the maximum 2-height of all codewords.* □

Let the *minimum distance* of the code $\mathcal{C}$

$$d(\mathcal{C})$$

be the smallest positive integer such that there exists a codeword $\mathbf{c} = (c_0, c_1, \cdots, c_{n-1}) \in \mathcal{C}$ that has exactly $d(\mathcal{C})$ non-zero numbers (out of its n numbers). That is, if π is its Ranking Permutation, then

$$|c_{\pi(0)}| \geq |c_{\pi(1)}| \geq \cdots \geq |c_{\pi(d(\mathcal{C})-1)}| > |c_{\pi(d(\mathcal{C}))}| = |c_{\pi(d(\mathcal{C})+1)}| = \cdots = |c_{\pi(n-1)}| = 0$$

So for that codeword $\mathbf{c}$, we have

$$1 = h_0(\mathbf{c}) \leq h_1(\mathbf{c}) \leq \cdots \leq h_{d(\mathcal{C})-1}(\mathbf{c}) < h_{d(\mathcal{C})}(\mathbf{c}) = h_{d(\mathcal{C})+1}(\mathbf{c}) = \cdots = h_{n-1}(\mathbf{c}) = \infty$$

Therefore for the Analog Code $\mathcal{C}$, we have

$$1 = h_0(\mathcal{C}) \leq h_1(\mathcal{C}) \leq \cdots \leq h_{d(\mathcal{C})-1}(\mathcal{C}) < h_{d(\mathcal{C})}(\mathcal{C}) = h_{d(\mathcal{C})+1}(\mathcal{C}) = \cdots = h_{n-1}(\mathcal{C}) = \infty$$

It is known that $d(\mathcal{C}) \leq n - k + 1$ (by a result known as the Singleton bound). Therefore even if we do not know what $d(\mathcal{C})$ is, we still know that

$$h_{n-k+1}(\mathcal{C}) = h_{n-k+2}(\mathcal{C}) = \cdots = h_{n-1}(\mathcal{C}) = \infty$$

2 Definition of the "m-Height Problem"

Given a $k \times n$ generator matrix G and a positive integer $m \leq n - k$, we would like to compute the m-height of the corresponding Analog Code $\mathcal{C}$. We call it the **m-Height Problem**. It is defined as follows.

m-Height Problem

Input: The problem has the following inputs:

1. Positive integers k, n and m, where $k < n$ and $m \leq n - k$.
2. A $k \times n$ real-valued full-rank generator matrix

$$G = \begin{bmatrix} g_{0,0} & g_{0,1} & \cdots & g_{0,n-1} \\ g_{1,0} & g_{1,1} & \cdots & g_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ g_{k-1,0} & g_{k-1,1} & \cdots & g_{k-1,n-1} \end{bmatrix}$$

   where no column is the all-zero vector.

3. An Analog Code

$$\mathcal{C} = \{\mathbf{x}G \mid \mathbf{x} \in \mathbb{R}^k\}$$

Output: the m-height of $\mathcal{C}$.

3 LP-based Algorithm for m-Height Problem

In this section, we present an algorithm that solves the "m-height problem" based on linear programming (LP). (The proof for the algorithm has been shown in a published research paper.)

Let $[n\rangle$ denote the integer set $\{0, 1, \cdots, n-1\}$, namely

$$[n\rangle = \{0, 1, \cdots, n-1\} \text{ for any positive integer } n.$$

Let $\mathcal{S}$ denote a subset of $[n\rangle$ with m distinct integers, namely, $\mathcal{S} \subseteq [n\rangle$ and $|\mathcal{S}| = m$. Let $\bar{\mathcal{S}} = [n\rangle \setminus \mathcal{S}$ be the subset of integers that are in $[n\rangle$ but not in $\mathcal{S}$. Let $\Lambda(n, m)$ denote the set of all possible values for $\mathcal{S}$, namely, the set of all subsets of $[n\rangle$ of cardinality m.

Example 7. *Let $n = 5$ and $m = 2$. Then $[n\rangle = \{0, 1, 2, 3, 4\}$, and*

$$\Lambda(n, m) = \{\{0,1\}, \{0,2\}, \{0,3\}, \{0,4\}, \{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}, \{3,4\}\}.$$

When $\mathcal{S} = \{0, 1\} \in \Lambda(n, m)$, we have $\bar{\mathcal{S}} = \{2, 3, 4\}$. When $\mathcal{S} = \{2, 4\} \in \Lambda(n, m)$, we have $\bar{\mathcal{S}} = \{0, 1, 3\}$.

Given any set $\mathcal{S} \in \Lambda(n, m)$ and any integer $j \in \mathcal{S}$, let

$$LP_{\mathcal{S},j}$$

denote the following linear program with k real-valued variables u_0, u_1, $\cdots$, u_{k-1}:

$$\begin{array}{ll} \text{maximize} & \sum_{i\in[k\rangle} g_{i,j} \cdot u_i \\ \text{s.t.} & -1 \le \sum_{i\in[k\rangle} g_{i,t} \cdot u_i \le 1 \quad \text{for } t \in \bar{\mathcal{S}} \end{array}$$

Let $z_{\mathcal{S},j}$ denote the optimal objective value of the linear program $LP_{\mathcal{S},j}$, namely, the maximum value of the objective function $\sum_{i\in[k\rangle} g_{i,j} \cdot u_i = g_{0,j}u_0 + g_{1,j}u_1 + \cdots + g_{k-1,j}u_{k-1}$ found by the linear program. Then for the Analog Code $\mathcal{C}$, its m-height is

$$h_m(\mathcal{C}) = \max_{\mathcal{S}\in\Lambda(n,m),\ j\in\mathcal{S}} z_{\mathcal{S},j}$$

Example 8. *Following our previous examples, let $k = 2$, $n = 5$, and $m = 2$. Let the generator matrix G be*

$$G = \begin{bmatrix} 1 & 0 & 0.4759809 & 0.9938236 & 0.819425 \\ 0 & 1 & -0.8960798 & -0.7442706 & 0.3345122 \end{bmatrix}.$$

Since $n = 5$ and $m = 2$, we have

$$\Lambda(5, 2) = \big\{\{0,1\}, \{0,2\}, \{0,3\}, \{0,4\}, \{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}, \{3,4\}\big\}.$$

Also, the equation

$$h_m(\mathcal{C}) = \max_{\mathcal{S}\in\Lambda(n,m),\, j\in\mathcal{S}} z_{\mathcal{S},j}$$

becomes

$$h_2(\mathcal{C}) = \max_{\mathcal{S}\in\Lambda(5,2),\, j\in\mathcal{S}} z_{\mathcal{S},j}.$$

Let $\mathcal{S} = \{1, 3\}$ and $j = 3 \in \mathcal{S}$. Then

$$\bar{\mathcal{S}} = \{0, 2, 4\}.$$

Therefore, the linear program $LP_{\mathcal{S},j}$, namely $LP_{\{1,3\},3}$, is

$$\begin{array}{ll} \textit{maximize} & 0.9938236\, u_0 - 0.7442706\, u_1 \\ \textit{subject to} & -1 \le u_0 \le 1, \\ & -1 \le 0.4759809\, u_0 - 0.8960798\, u_1 \le 1, \\ & -1 \le 0.819425\, u_0 + 0.3345122\, u_1 \le 1. \end{array}$$

□

Since the m-height $h_m(\mathcal{C})$ increases with m, the above linear-program based method naturally leads to an algorithm that computes all the m-height of a code $\mathcal{C}$, for $m = 0, 1, \cdots, n-1$, and also discovers the minimum distance $d(\mathcal{C})$:

1. $h_0(\mathcal{C}) = 1$.
2. For $m = 1, 2, 3 \cdots$, compute
$$h_m(\mathcal{C}) = \max_{\mathcal{S} \in \Lambda(n,m),\ j \in \mathcal{S}} z_{\mathcal{S},j}$$
by solving the linear programs $LP_{\mathcal{S},j}$ for all $\mathcal{S} \in \Lambda(n,m)$ and $j \in \mathcal{S}$. Stop as soon as we meet the first value m^* such that $h_{m^*}(\mathcal{C}) = \infty$. We get $d(\mathcal{C}) = m^*$.
3. For $m = m^* + 1, m^* + 2, \cdots, n-1$, $h_m(\mathcal{C}) = \infty$.

The above algorithm solves $m\binom{n}{m}$ linear programs of k variables to compute an $h_m(\mathcal{C})$.

4 Your Task

Your task for the project is to design and train a deep neural network (DNN) that efficiently estimates the m-height of an Analog Code $\mathcal{C}$, given its generator matrix G and the three parameters n, k and m. The standardized input to your code should be:

- The first input is an integer n.
- The second input is an integer k.
- The third input is an integer m.
- The fourth input is a $k \times (n-k)$ matrix $P_{k\times(n-k)}$ (as a 2-dimensional numpy array), which forms the last $n-k$ columns of a systematic $k \times n$ generator matrix G.

The output of your code should be a prediction of the m-height (of the Analog Code whose generator matrix is the systematic generator matrix $G = [I_{k\times k}|P_{k\times(n-k)}]$), which is a real number greater than or equal to 1.

Note that your code should focus on deep learning, namely, it should purely use deep neural network(s) to estimate the m-height. Any additional codes other than the deep neural network(s) — whether it is the preprocessing code that prepares the input data to the neural networks (such as normalizing the input data), or the postprocessing code that turns the output of the neural networks to the final output data of the correct format (such as converting the "logarithm of the predicted m-height" to m-height), or any additional code that handles exceptions (such as making sure the predicted m-height is at least 1, handling the divide-by-zero error, handling the out-of-range error, etc.) — should all be kept to minimal (otherwise the project will receive no grade).

As an example, operations that directly use a function to compute special features of a matrix — such as torch.linalg.svdvals, etc. — cannot be used (even inside a neural network). (If they are used, they will be considered pre-processing/post-processing codes, but that would be a violation of the project's requirement.) That is because such operations are not layers of deep neural networks. For example, torch.linalg.svdvals is a linear algebra function, not a neural network layer. It computes the singular values of a matrix, while deep learning layers are things like nn.Linear, nn.Conv2d, or nn.ReLU that are used inside a model's forward/backward pass. In other words, operations that are not neural network layers with trainable weights (except for very common layers such as SoftMax, Layer Normalization, etc.) are usually not allowed in the project.

If the requirement described in the two paragraphs above is violated, the whole project will receive no point.

Your code can contain one DNN (namely, you use one DNN to handle all possible values of n, k, m and G) or multiple DNNs (e.g., you train a separate DNN for each specific tuple (n, k, m), and select the suitable

DNN to provide the answer for every input (n, k, m, G), like a "mixture-of-experts (MoE)" model). Your model should run substantially faster than the LP-based algorithm introduced above.

In this project, we will focus on the following values of n, k and m:

$$n = 9, \quad k \in \{4, 5, 6\}, \quad m \in \{2, 3, \cdots, n-k\}$$

They include the following parameters:

1. $n = 9$, $k = 4$, $m = 2$.
2. $n = 9$, $k = 4$, $m = 3$.
3. $n = 9$, $k = 4$, $m = 4$.
4. $n = 9$, $k = 4$, $m = 5$.
5. $n = 9$, $k = 5$, $m = 2$.
6. $n = 9$, $k = 5$, $m = 3$.
7. $n = 9$, $k = 5$, $m = 4$.
8. $n = 9$, $k = 6$, $m = 2$.
9. $n = 9$, $k = 6$, $m = 3$.

So when you design your DNN-based model, you can just focus on the above parameters (instead of more general values).

What you should submit:

1. Your trained DNN (or DNNs) for the task.
2. A complete Jupyter notebook that can run readily in Google CoLab, including:
   (a) All the modules (i.e., software packages) that need to be installed in Google Colab in order to run your code.
   (b) All your codes, include how you built the DNN, how you trained your DNN, what the training performance was, how you tested your DNN, and what the test performance was. All these parts should be accompanied with clear and detailed explanations (as documentation for the codes) using the "text cells" in the Jupyter notebook (which you can put above the corresponding "code cells").
   (c) The final cell of the Jupyter notebook should be a cell that we can run to test your DNN-based model. It should be able to load the DNN you have trained, allow us to input a test dataset (where each input sample is a tuple $(n, k, m, P_{k\times(n-k)})$), and outputs the list of predicted m-heights. (How we measure the performance of your predicted m-heights will be introduced below.)

Note: after submitting your files, please double check to make sure they are the correct files. If incorrect files are submitted, the project will receive **no point**. If correct files are submitted but they do not fully satisfy the requirements above, points will be deducted.

Note: We provide a training set of two files – an input file that contains samples of (n, k, m, P) and a corresponding output file that contains the corresponding m-heights – for the project. The test set files will be of the same format. Your code should be able to take the input file as input and return a file of the same format as the output file WITHOUT any modification, otherwise the project will receive **no point**. The only exception is that if the input file is too large, you are allowed to partition it into a few smaller files, get the corresponding output files, and assemble them into a final output file.

The performance of your solution will be measured as follows. Given a sample $(n, k, m, P_{k\times(n-k)})$, let

$$y \in [1, \infty)$$

be the **true** m**-height** of the Analog Code (whose generator matrix is $G = [I_{k\times k}|P_{k\times(n-k)}]$), and let

$$\hat{y} \in [1, \infty)$$

be the m-height **predicted** by your DNN-based model. (For simplicity, in the test set that will be used to test your code, the true m-height will always be a finite number, not infinity. So your prediction $\hat{y}$ should always be a finite real number, too. Also, make sure that your $\hat{y}$ is always at least 1, because that is a basic property of m-height that should never be violated. Otherwise – that is, if ANY predicted or submitted m-height is less than 1 or is infinity – the whole project will receive **no point**.) The **cost** of your prediction is defined to be

$$\sigma(y, \hat{y}) \triangleq (\log_2 y - \log_2 \hat{y})^2$$

Note that the cost reaches its minimum value 0 if and only if $y = \hat{y}$.

Given n, k and m, let

$$T_{n,k,m} \ = \ \{G^{[t]}_{n,k,m} \mid t \in [N_{n,k,m}\rangle\}$$

be a set of $N_{n,k,m}$ generator matrices, where for $t \in [N_{n,k,m}\rangle$, the matrix $G^{[t]}_{n,k,m}$ is a randomly sampled (but not necessarily *uniformly* randomly sampled) $k \times n$ **systematic** generator matrix (namely, its first k columns form a $k \times k$ identity matrix), where no column is an all-zero vector and whose m-height is finite. Furthermore, **each number in the last** $n-k$ **columns of** $G^{[t]}_{n,k,m}$ **is between -100 and 100**. Let $y^{[t]}_{n,k,m}$ be the true m-height of the corresponding Analog Code, and let $\hat{y}^{[t]}_{n,k,m}$ be the m-height predicted by your DNN-based model. Let

$$\sigma^{[t]}_{n,k,m} \triangleq \sigma(y^{[t]}_{n,k,m}, \hat{y}^{[t]}_{n,k,m}) = (\log_2 y^{[t]}_{n,k,m} - \log_2 \hat{y}^{[t]}_{n.k,m})^2$$

be the corresponding cost. The **average cost** for the set $T_{n,k,m}$ is defined to be

$$\sigma_{n,k,m} \triangleq \frac{1}{N_{n,k,m}} \sum_{t\in[N_{n,k,m}\rangle} \sigma^{[t]}_{n,k,m}$$

5 Method of Grading

For each parameter set (n, k, m) (such as $n = 9$, $k = 4$, $m = 2$), we will use a randomly sampled test set $T_{n,k,m}$ to get the *average cost*

$$\sigma_{n,k,m} \ \geq \ 0$$

of your submitted DNN-based model. Let $\theta_{n,k,m}$ and $\alpha_{n,k,m}$ be two positive real parameters. The **score** of your model for this parameter set (n, k, m) is set as

$$\beta_{n,k,m} \triangleq \max\{(\frac{100}{(e^{\alpha_{n,k,m}(\sigma_{n,k,m}-\theta_{n,k,m})}+1)} - 50) \cdot \frac{50}{\frac{100}{(e^{-\alpha_{n,k,m}\cdot\theta_{n,k,m}}+1)} - 50} + 50, 0\}$$

The two parameters $\theta_{n,k,m}$ and $\alpha_{n,k,m}$ are chosen in the following way:

- Say that there are totally M submissions, whose scores are sorted as

  $$\sigma^{(1)}_{n,k,m} \leq \sigma^{(2)}_{n,k,m} \leq \cdots \leq \sigma^{(M)}_{n,k,m}$$

  where $\sigma^{(j)}_{n,k,m}$ is the jth student's *average score* for $j = 1, 2, \cdots, M$. Then

  $$\theta_{n,k,m} = \sigma^{(\lceil M/2\rceil)}_{n,k,m}$$

  and $\alpha_{n,k,m}$ is chosen to be the value such that

  $$e^{\alpha_{n,k,m}(\sigma^{(\lceil M/4\rceil)}_{n,k,m}-\theta_{n,k,m})} = \frac{1}{3}$$

namely,

$$\alpha_{n,k,m} = \frac{\ln 3}{\sigma_{n,k,m}^{(\lceil M/2 \rceil)} - \sigma_{n,k,m}^{(\lceil M/4 \rceil)}}$$

Note: the above "score curve" $\beta_{n,k,m}$ has a seemingly complex formula, but it is actually a simple sigmoid function, except that we flip it from left to right so that the smaller the cost $\sigma_{n,k,m}$ is, the greater the score $\beta_{n,k,m}$ is. (If $\sigma_{n,k,m}$ is 0, then $\beta_{n,k,m}$ is 100.) The max function is used in the formula to ensure that the score is at least 0.

Let S denote the set of all values for the parameter set (n, k, m). (For example, with $n = 9$, $k \in \{4, 5, 6\}$ and $m \in \{2, 3, \cdots, n - k\}$, the set S has 9 values for the parameter set (n, k, m).) Then the **score** of your DNN-based model is set as

$$\text{SCORE}_{\text{model}} \triangleq \frac{1}{|S|} \sum_{(n,k,m) \in S} \beta_{n,k,m} \in [0, 100]$$

which is the average score over the different parameter settings in S.

The DNN-based model accounts for 90% of the final grade of your submission. The other 10% is based on the code and explanations (namely, documentation) in your Jupyter notebook, including (but not limited to) if the codes are complete, if the explanations are clear and detailed, if there is no issue in running your code, etc. Let

$$\text{SCORE}_{\text{notebook}} \in [0, 100]$$

denote the score for your Jupyter notebook. Then the **final grade** for your submission is

$$\text{SCORE}_{\text{model}} \times 90\% \ + \ \text{SCORE}_{\text{notebook}} \times 10\%$$